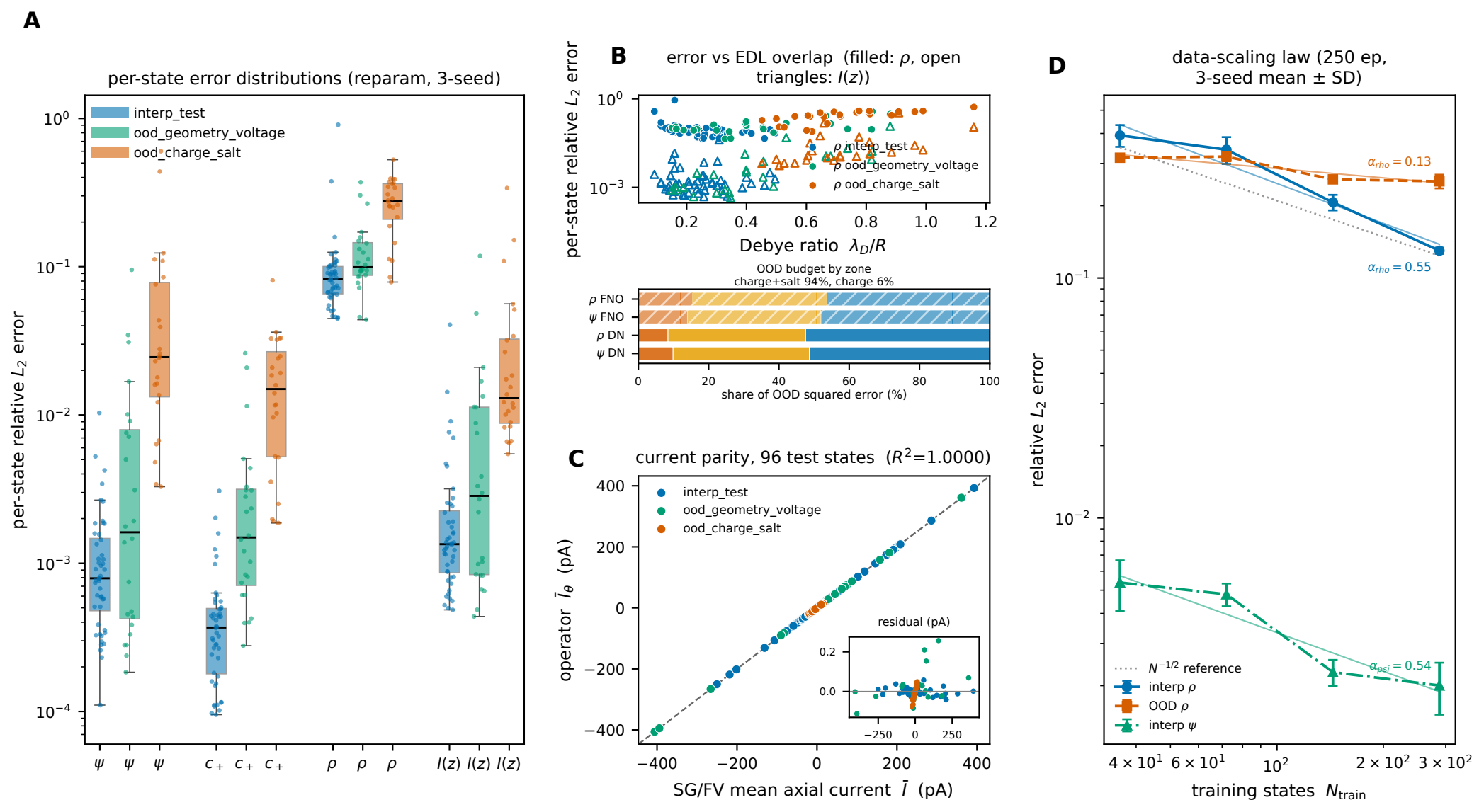

A
per-state error distributions (reparam, 3-seed)
interp_test
ood_geometry_voltage
ood_charge_salt
per-state relative $L_2$ error
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$\psi$ $\psi$ $\psi$ $c_+$ $c_+$ $c_+$ $\rho$ $\rho$ $\rho$ $I(z)$ $I(z)$ $I(z)$
B
error vs EDL overlap (filled: $\rho$, open triangles: $I(z)$)
per-state relative $L_2$ error
$10^{0}$
$10^{-3}$
$\rho$ interp_test
$\rho$ ood_geometry_voltage
$\rho$ ood_charge_salt
0.2 0.4 0.6 0.8 1.0 1.2
Debye ratio $\lambda_D/R$
OOD budget by zone
charge+salt 94%, charge 6%
$\rho$ FNO
$\psi$ FNO
$\rho$ DN
$\psi$ DN
0 20 40 60 80 100
share of OOD squared error (%)
C
current parity, 96 test states ($R^2$=1.0000)
interp_test
ood_geometry_voltage
ood_charge_salt
operator $\bar{I}_\theta$ (pA)
400 200 0 −200 −400
residual (pA)
0.2
0.0
−250 0 250
−400 −200 0 200 400
SG/FV mean axial current $\bar{I}$ (pA)
D
data-scaling law (250 ep, 3-seed mean ± SD)
relative $L_2$ error
$10^{-1}$
$10^{-2}$
$\alpha_{rho} = 0.13$
$\alpha_{rho} = 0.55$
$\alpha_{psi} = 0.54$
$N^{-1/2}$ reference
interp $\rho$
OOD $\rho$
interp $\psi$
$4 \times 10^{1}$ $6 \times 10^{1}$ $10^{2}$ $2 \times 10^{2}$ $3 \times 10^{2}$
training states $N_{\text{train}}$